

MULTIPLE STARS

Most of the stars in the Milky Way are members of either binary or multiple systems—single stars like the Sun are more unusual. These systems vary from distant pairs in slow, centuries-long orbits around a common center of mass to tightly bound groups that orbit each other in days

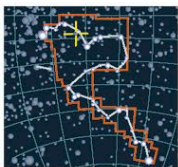
and may even distort each other's shape. Most multiples are so close together that we know about them only from their spectra. They also vary widely in size and color—stars of any age and type can be members of a multiple star system.

TRAPEZIUM

The multiple star known as Theta Orionis, or the Trapezium, is a system containing at least 10 individual stars.

TRIPLE STAR

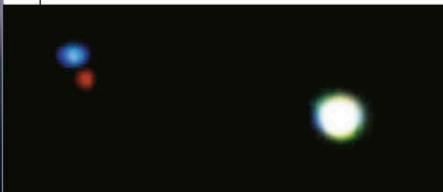
Omicron Eridani



ERIDANUS

DISTANCE FROM SUN
16 light-years
MAGNITUDE 9.5
SPECTRAL TYPE DA

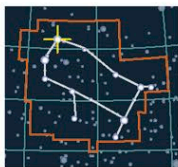
Originally Omicron (o) Eridani was classed as a double star, Omicron-1 Eridani and Omicron-2 Eridani. Nineteenth-century observations revealed that the system is actually three stars, now called 40 Eridani A, B, and C. A is a main-sequence orange-red dwarf, and C is a faint red dwarf. However, it is 40 Eridani B that is the gem. This young white dwarf is the brightest white dwarf visible through a small telescope.



TRIPLE SYSTEM

SEXTUPLE SYSTEM

Castor



GEMINI

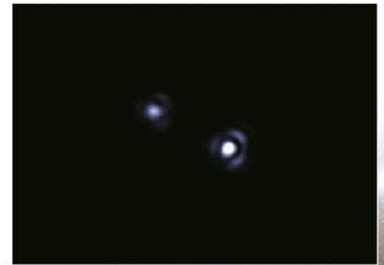
DISTANCE FROM SUN
51 light-years
MAGNITUDE 1.6
SPECTRAL TYPE A2

Easily visible to the naked eye, Castor appears to be an ordinary A-type star. However, a telescope reveals that Castor is in fact a pair of bright A-type stars, Castor A and Castor B, with a fainter third companion, Castor C. Spectrographic analysis shows that both the A and B components of Castor are themselves double stars.

Castor A consists of two stars in a very close 9.2-day orbit, while Castor B's components orbit each other in a rapid 2.9 days. The faint Castor C star is also a double—a pair of faint red-dwarf stars orbiting each other with a period of only 20 hours. Castor is therefore a sextuplet star: a double-double-double.

DOUBLE-DOUBLE-DOUBLE

Castor (boxed) and its neighbor Pollux are the two brightest stars in Gemini (below). Only when viewed through a telescope are the individual stars, Castor A and Castor B, separated (right).



QUADRUPLE STAR

Algol



PERSEUS

DISTANCE FROM SUN
93 light-years
MAGNITUDE 2.1
SPECTRAL TYPE B8

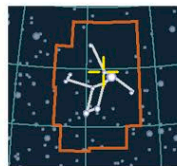
Algol, or Beta (β) Persei, appears to the naked eye as a single star. However, exactly every 2.867 days, the star's brightness drops by 70 percent for a few hours—a variation that was discovered as early as 1667. This variation is caused by Algol's being eclipsed by a faint giant star Algol B, which is larger than the bright primary Algol A.



ECLIPSING BINARY

QUADRUPLE STAR

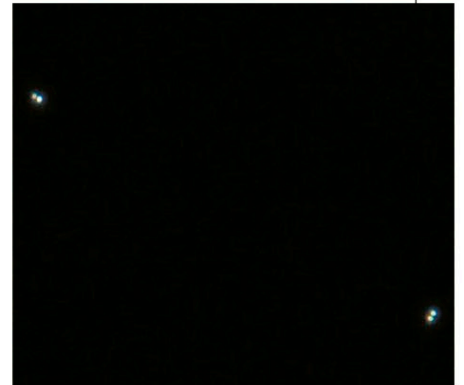
Epsilon Lyrae



LYRA

DISTANCE FROM SUN
160 light-years
MAGNITUDE 3.9
SPECTRAL TYPE A4

Epsilon (ε) Lyrae is visible as a double star on a clear, dark night, but closer observation reveals that, in fact, each star is itself a double. Unlike other double-double systems, Epsilon Lyrae is within reach of amateur astronomers—its four component stars can each be seen through a telescope, and spectroscopy is not needed to detect their presence (see p.274). The two bright stars visible to the naked eye, Epsilon-1 and Epsilon-2, are widely separated, with an orbital period of millions of years. The components of each pair



ISOLATED PAIRS

This double-double system is easily separated into its four components through a telescope. Although the stars in each pair are strongly bound to one another, the link between the pairs is tenuous.

orbit much more closely, with a period of about 1,000 years. Epsilon-1 and Epsilon-2 are so far apart, they are hardly bound by gravity at all, and eventually Epsilon Lyrae will become two separate star systems.

QUADRUPLE STAR

Mizar and Alcor



URSA MAJOR

DISTANCE FROM SUN
81 light-years
MAGNITUDE 2
SPECTRAL TYPE A2

Although Mizar and Alcor are a famous naked-eye double, easily visible in the handle of the Big Dipper and known since ancient times as the horse and rider, it is still unknown whether or not they are a genuine double. Mizar itself is a double star—the first double star to be discovered. Spectrography reveals, however, that Mizar is a double-double star—that is, two double stars in orbit around each other.

MIZAR THROUGH A TELESCOPE

